PRODUCT USER MANUAL

Atlantic - European North West Shelf - Ocean Physics Analysis and Forecast Product

NWSHELF_ANALYSISFORECAST_PHY_004_013

Issue: 3.0

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2020/08	All	Merged PUM_004_013 and 004_014. Updated for Dec 2020 EIS.	M. Tonani	
1.1	2021/01	All	Updated for May	I. Ascione	
1.2	2021/08	II.3 and II.4	Details on vertical grid and vertical interpolation	M. Tonani and I. Ascione	
1.3	2022/01	II.2 and II.4	Upgrade to GLO-HR boundary conditions and new MDT	A. Saulter and I. Ascione	C. Derval
2.0	2023/06/16	All	**[EIS September 2023]** First release of the NWS ocean circulation analysis and forecast product, as an extraction of the IBI operational system.	R. Aznar, A. Amo, L. Castrillo, G. Reffray, R. Escudier	S. Cailleau, M. G. Sotillo
2.1	2024/05/31	All	**[EIS November 2024]** New variables distributed: Detided surface currents and mean sea level, and vertical velocity. Forecast horizon increased from 5 to 10 days.	R. Aznar, L. Castrillo, G. Reffray, R. Escudier	S. Cailleau, M. G. Sotillo
3.0	2025/05	All	**[EIS November 2025]** Re-introduction of the NWS physical analysis and forecasting product by the UK Met Office	J. Tinker, T. Brüning	Copernicus Marine Product Management

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
NWS	North-West Shelf
TAC	Thematic Assembly Centre
GLO	Global
IOC	Intergovernmental Oceanographic Commission of UNESCO
NRT	Near Real Time
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	A hindcast done without data assimilation.
ANALYSIS	A hindcast adjusted with data assimilation
SSS	Sea Surface Salinity
SSC	Sea Surface Current
SSH	Sea Surface Height

SST	Sea Surface Temperature
Zonal Velocity	West to East component of the horizontal velocity vector
Meridional Velocity	South to North component of the horizontal velocity vector

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us.

1) INTRODUCTION

a) Summary

This document is the user manual for the North West-European Shelf analysis and forecast product:

NWSHELF_ANALYSISFORECAST_PHY_004_013

This product is generated by a coupled physical-wave forecasting system. The corresponding wave product is **NWSHELF_ANALYSISFORECAST_WAV_004_014**, which is described here.

Every day:

- the time series (analysis for 2 days before present and 7 days forecast) is updated at the end of the operational production process
- old forecasts are deleted from the MDS every day

The product is published on and available through the MDS after automatic and human quality controls. Downloaded files are in NetCDF format and follow CF-1.7 convention.

The analysis and forecasting system is described in the Quality Information Document (QUID):

https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-013.pdf

More detailed information can be obtained from the Copernicus Marine Service Team.

b) History of changes

Date	Description of changes and impacted product
July 2020	Correction of 30 min shift in time record of hourly dataset.
December 2020	Removal of TMB dataset Addition of SST, SSS, SSC, MLD, BED hourly datasets Addition of SSC and SSH 15min datasets time units in seconds since 1970-01-01T00:00:00Z
February 2022	Boundary conditions have been upgraded
November 2023	First release of the NWS ocean circulation analysis and forecast product, as an extraction of the IBI operational system.
November 2024	New variables distributed: Detided surface currents and mean sea level, and vertical velocity. Forecast horizon increased from 5 to 10 days.
November 2025	Re-introduction of the NWS physical analysis and forecasting product by the UK Met Office including a 7-day forecast

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

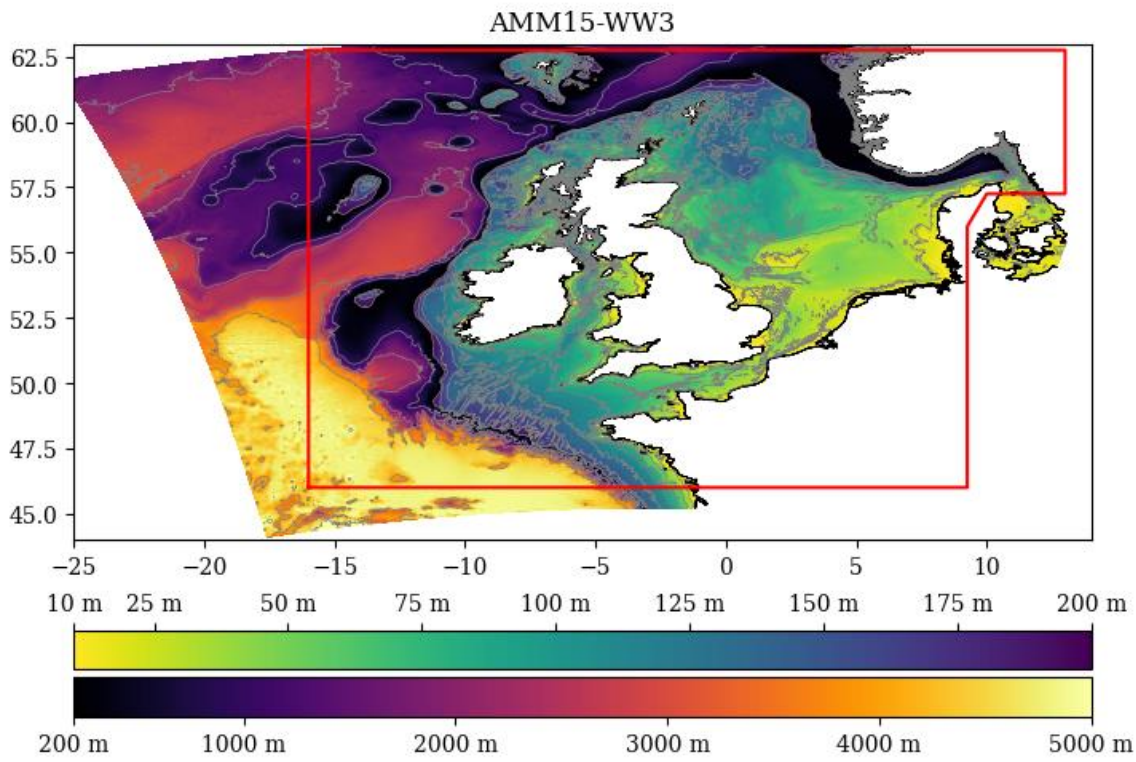
Product Lines	NORTHWESTSHELF_ANALYSIS_FORECAST_PHY_004_013	
Geographical coverage	16°W to 13°E; 46°N to 62.75°N. Baltic boundary: all the grid points in the Baltic and the Kattegat South of 57.25°N are masked.	
Variables	• Potential temperature • Salinity • Sea surface height • Horizontal velocity (eastward and northward components) • Bottom Temperature • Mixed Layer depth	
Product Type	Near Real Time - Analysis	Near Real Time - Forecast
Available time series	2 years before present	7 days forecast
Temporal resolution	Daily means (25-hour mean[1]); Hourly & Quarter-hourly instantaneous	Daily means (25-hour mean[1]); Hourly & Quarter-hourly instantaneous
Temporal resolution WAV_004_014	Hourly instantaneous	Hourly instantaneous
Update Frequency	Daily	Daily
Target delivery time	daily at 12 pm (noon) UTC	
Horizontal resolution	1.9± 0.4 km longitude x 1.5km latitude 1/33° longitude x 1/74° latitude	
Number of vertical levels	33	
Format	NetCDF4 – CF1.7	
Delivery mechanism	Copernicus Marine Toolbox	

[1] Our daily mean products are the mean of 25 instantaneous means, including the previous and following midnight values – this effectively removes the effect of the tide in the data.

b) Production System Description

The operational North-West-European Shelf analysis and forecast products are generated by a coupled physical-wave forecasting system at 1/33 degree (longitude) and 1/74° (latitude). It provides a 7-day forecast and two analysis days of 3D (PHY_004_013) and surface (WAV_004_014) updated daily. This product (the physical product) includes detided daily mean, and hourly and quarter hourly instantaneous files. A rolling archive of 2 years of analysis is available.

NORTH WEST EUROPEAN SHELF (16°W - 13°E; 46°N - 63°N)
1/33° longitude x 1/74° latitude (958 x 1240 grid points)

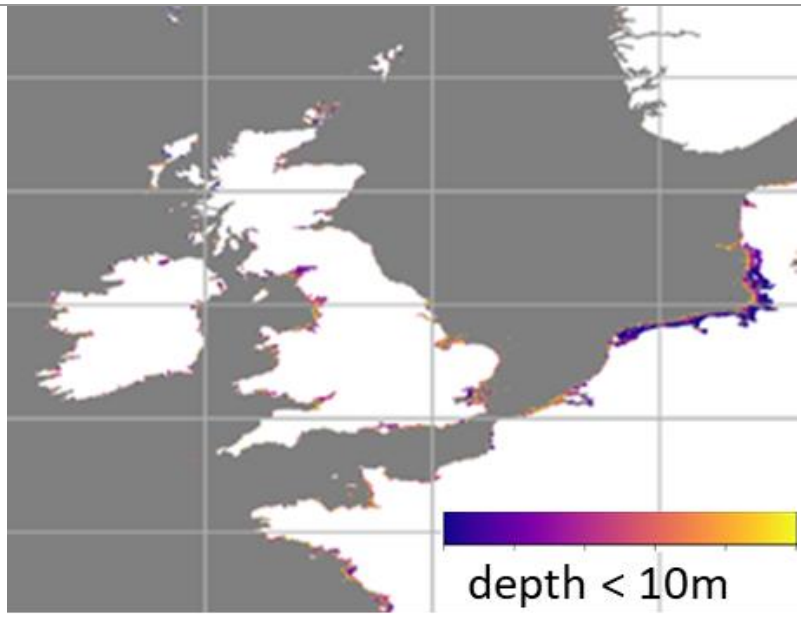


NWS bathymetry (m) of the AMM15 model domain, with the red line showing the product domain.

The domain extends beyond the continental shelf in order to place the model's boundary region in the deep waters of the adjacent North-East Atlantic, but the focus region for the model comprises open waters of the shelf seas, i.e. (using UK terminology) the North Sea, Irish Sea, English Channel, Celtic Sea and Bay of Biscay.

The model is a nested regional ocean model configuration, coupled with a wave model. PHY and WAV products are delivered on the same horizontal grid.

A **minimum depth of 10m** is imposed in the ocean model for considering the sea level tidal excursion. Products in areas where the bathymetry is less than 10m could be affected by this modelling choice.



1) What is done:

We take every bathymetric grid point < 10 m in the domain and reassign it with a depth of 10 m, coloured point in the figure above. These grid points are interpolated back to their original depth before delivering the file to the CMEMS catalogue. Areas such as the Wadden Sea are particularly affected (see below reference for a case study in the Wadden sea).

2) Why it is done:

This is because version 3.6 of NEMO does not have wetting and drying capability. If the bathymetry were not modified, it would mean that with the large tides that exist in this domain the model would not be able to handle the case when the tide goes out.
The minimum depth prevents such a situation from occurring and keeps the model as a whole stable.

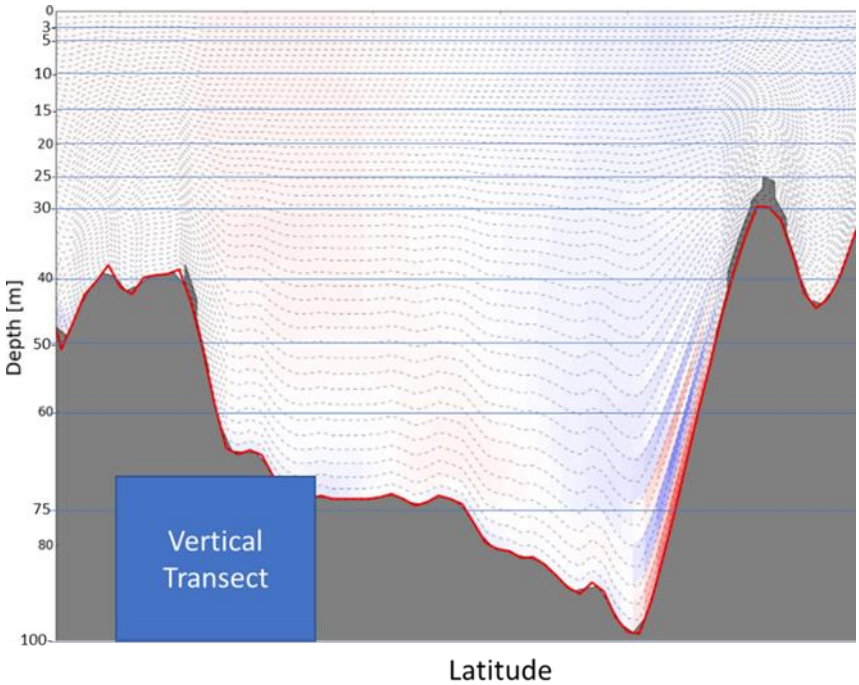
3) What is the likely effect.

The prorogation of tides and surges are inherently related to the water depth. The speed is proportional to $\sqrt{gH}$ where H is the water depth and thus as the model water depth is deeper than reality the prorogation will be incorrect.
In cases where the seabed is exposed then this process is completely absent from the model by design (there will be water in the model where in reality there is none). It is thus not recommended to the user to directly use the model fields in regions where there are extensive areas of bathymetry less than 10 m in depth. If a model is to be nested within this product, then it is recommended that the boundaries of the nested domain are sufficiently far from such areas to reduce their impact.

4) Wetting and Drying is implemented in NEMO 4.

This specific problem is addressed with the wetting and drying module in NEMO 4 which will be part of future development of this operational Copernicus product.
The interested reader is directed to the recent paper on its development:
https://doi.org/10.1016/j.ocemod.2020.101708
Enda O'Dea, Michael J. Bell, Andrew Coward, Jason Holt, Implementation and assessment of a flux limiter based wetting and drying scheme in NEMO, Ocean Modelling, Volume 155,2020.

Horizontal grid	1.9± 0.4 km longitude x 1.5km latitude The longitude grid resolution varies with latitude. It is coarser at lower latitude (lat=46°N ->1.3 km), and finer at higher latitude (lat=63°N -

	>1.5km). The latitudinal grid resolution is constant in the whole model domain and is 1.5km). It should be noted that whilst the North-West Shelf physical product is provided on a regular grid at 1/33° longitude x 1/74° latitude, this is derived from a model with a native grid using a resolution of 1/60°. An Euler rotation is applied to the spherical-polar grid such that a pseudo equator passes through the domain, resulting in a grid with a regular resolution of approximately 1.8 km both zonally and meridionally (Guihou et al. 2017). The wave model native horizontal grid is a Spherical Multiple Cell grid with a variable resolution of 3-1.5 km, designed for improved resolution of islands and coastlines.
Vertical grid	The products are interpolated for users' convenience, into 33 standard IOC geopotential levels: [surface, 3, 5, 10, 15, 20, 25, 30, 40, 50, 60, 75, 100, 125, 150, 175, 200, 225, 250, 300, 350, 400, 450, 500, 550, 600, 750, 1000, 1500, 2000, 3000, 4000, 5000 m]. The surface level is not interpolated, it is the first model level. It has a thickness of 1m where the bathymetry is deeper than 50m. Its thickness is less than 1m where the bathymetry is shallower than 50m.  *Vertical transect: it shows the differences between the model native vertical grid (dotted line) and the CMEMS products interpolated at standard geopotential depths (blue lines). The model bathymetry (red lines) doesn't match exactly the bathymetry (shaded grey area) due to numerical constrains.* The model native vertical grid has Hybrid S-σ-z-coordinates. The terrain-following coordinates system uses a stretching function to maintain near uniform vertical resolution at the surface. The number of S-σ-z levels is 51 almost everywhere in model domain. It means that the thickness of the model vertical grid cells is smaller in shallow area compared to the deep part of the basin.

	The thickness of the model native vertical grid varies from 0.3m (in areas with a bathymetry < 50m) to 99m (where the bathymetry is > 4000m). The bottom temperature (bottomT) is output directly from the native model, it is not interpolated. All other CMEMS NWS-MFC products are vertically interpolated from the 51 native terrain following coordinates to fixed depths.
System Name	AMM15
Model version	NEMO3.6 (Madec et al., 2017) coupled with wave model WAVEWATCH III v 4.18
Coupling	2-way on-line ocean-wave coupling via OASIS3-MCT. • Modification of water-side surface stress on the ocean by wave growth and dissipation • Stokes-Coriolis force in the momentum equation • Wave-height-dependent ocean surface roughness The wave model is forced by the ocean surface currents. Coupling frequency: hourly.
Bathymetry	EMODnet 2015. In order to match the grid set-up in the hydrodynamic model, but improve the estimate of wave energy transmission in the vicinity of sub grid scale topographic features such as small islands and coastal headlands, a number of sub grid blocking cells are defined for the wave model following the method described by Chawla and Tolman (2008).
Initial conditions	The model run started on 10/01/2016 from initial condition from a long simulation.
Atmospheric forcings	Met Office Global Unified Model (MetUM)
Tidal constituents	11 tidal constituents: M_2, S_2, N_2, K_2, K_1, O_1, P_1, Q_1, M_4, MS_4, MN_4. Tidal forcing is included both on the open boundary conditions via Flather radiation boundary conditions (Flather, 1976) and through the inclusion of equilibrium tide. The Topex Poseidon cross-over solution (Egbert and Erofeeva, 2002; TPX07.2, Atlantic Ocean 2011-ATLAS) provides 11 constituents for amplitude and phase of surface height and velocity at 1/12° (Tonani et al. 2019).
Rivers	Daily climatology of gauge data averaged for 1980–2014. UK data were processed from raw data provided by the Environment Agency, the Scottish Environment Protection Agency, the Rivers Agency (Northern Ireland), and the National River Flow Archive (personal communication by Sonja M. van Leeuwen, CEFAS, 2016). For major rivers that were missing from this data set (e.g. along the French and Norwegian coast), data have been provided by the same climatology used by AMM7 (Vörösmarty et al., 2000 and Young and Holt, 2007).
Lateral boundaries	• Atlantic (T, S, SSH, barotropic u and v): Met Office Operational ORCA12 forecast implementation. • Baltic (T, S, barotropic u and v): BALTICSEA_ANALYSISFORECAST_PHY_003_006.
Data Assimilation scheme	3D Var FGAT. NEMOVAR.

Assimilated observations	• In situ and satellite (L2/L3 products) SST. • satellite SLA from CMEMS SL-TAC. • temperature and salinity subsurface profiles from GTS and CMEMS INS-TAC.

c) Dataset details

NORTHWESTSHELF_ANALYSIS_FORECAST_PHY_004_013	
Quarter hourly instantaneous (**PT15M-i**), Hourly Instantaneous (**PT1H-i**) and Daily means (**P1D-m**) fields of: • Potential temperature and salinity (3D) • Zonal, meridional and vertical velocity (3D), zonal and meridional barotropic velocity (2D) • Sea Surface Height (SSH), sea bottom temperature and 2D Mixed layer depth • Sea Surface Temperature (SST) • Sea Surface Salinity (SSS) • Sea Surface Current (SSC)	
Datasets	Variables name in the NetCDF file [Unit] Long_name CF Standard_name
cmems_mod_nws_phy-cur_anfc_1.5km-3D_PT1H-i	**vo** [ms^{-1}] Northward Current Velocity in the Water Column northward_sea_water_velocity
	uo [ms^{-1}] Eastward Current Velocity in the Water Column eastward_sea_water_velocity
	wo [ms^{-1}] Upward (vertical) Current Velocity in the Water Column upward_sea_water_velocity
	vbar [ms^{-1}] Barotropic Northward Current Velocity in the Water Column barotropic_northward_sea_water_velocity
	ubar [ms^{-1}] Barotropic Eastward Current Velocity in the Water Column barotropic_eastward_sea_water_velocity
cmems_mod_nws_phy-cur_anfc_1.5km-3D_P1D-m	**vo** [ms^{-1}] Northward Current Velocity in the Water Column northward_sea_water_velocity
	uo [ms^{-1}] Eastward Current Velocity in the Water Column eastward_sea_water_velocity
	wo [ms^{-1}] Upward (vertical) Current Velocity in the Water Column upward_sea_water_velocity
cmems_mod_nws_phy-tem_anfc_1.5km-3D_PT1H-i cmems_mod_nws_phy-tem_anfc_1.5km-3D_P1D-m	**thetao** [degrees_C] Sea Water Potential Temperature sea_water_potential_temperature

cmems_mod_nws_phy-sal_anfc_1.5km-3D_PT1H-i cmems_mod_nws_phy-sal_anfc_1.5km-3D_P1D-m	**so** [1e-3] Sea Water Salinity sea_water_salinity
cmems_mod_nws_phy-bottomt_anfc_1.5km-2D_PT1H-i cmems_mod_nws_phy-bottomt_anfc_1.5km-2D_P1D-m	**bottomT** [degrees_C] Sea floor potential temperature sea_water_potential_temperature_at_sea_floor
cmems_mod_nws_phy-mld_anfc_1.5km-2D_PT1H-i cmems_mod_nws_phy-mld_anfc_1.5km-2D_P1D-m	**mlotst** [m] Ocean mixed layer thickness defined by sigma theta (reference depth of 3m, dt_crit of -0.2) ocean_mixed_layer_thickness_defined_by_sigma_theta
cmems_mod_nws_phy-ssh_anfc_1.5km-2D_PT1H-i cmems_mod_nws_phy-ssh_anfc_1.5km-2D_PT15M-i	**zos** [m] Sea surface height above geoid sea_surface_height_above_geoid
cmems_mod_nws_phy-cur_anfc_1.5km-2D_PT1H-i cmems_mod_nws_phy-cur_anfc_1.5km-2D_PT15M-i	**vo** [ms^{-1}] Northward Current Velocity in the Water Column northward_sea_water_velocity
	uo [ms^{-1}] Eastward Current Velocity in the Water Column eastward_sea_water_velocity
cmems_mod_nws_phy-sss_anfc_1.5km-2D_PT1H-i	**so** [1e-3] Sea Water Salinity sea_water_salinity
cmems_mod_nws_phy-sst_anfc_1.5km-2D_PT1H-i	**thetao** [degrees_C] Sea Water Potential Temperature sea_water_potential_temperature
cmems_mod_nws_phy_anfc_1.5km_static	**mask** Land-sea mask sea_binary_mask
	depth [m] depth depth
	deptho [m] Bathymetry sea_floor_depth_below_geoid
	mdt [m] Mean dynamic topography sea_surface_height_above_geoid
	deptho_lev_interp deepest wet grid point of the interpolated 3D fields model_level_number_at_sea_floor

The model **bathymetry, Land-Sea mask** and Mean Dynamic Topography **(MDT)** of this product is also used by the product NWSHELF_ANALYSISFORECAST_WAV_004_014, as both products are produced by a coupled physical-wave system.

d) Additional information on some parameters

mlotst [m]	ocean_mixed_layer_thickness_defined_by_sigma_theta. It is the depth where the density increase compared to density at 10 m depth corresponds to a temperature decrease of 0.2°C in local surface conditions (θ10m, S10m, P0= 0 db, surface pressure)
zos [m]	sea_surface_height_above_geoid. The geoid is a surface of constant geopotential with which mean sea level would coincide if the ocean were at rest. The parameter "zos" is the difference between the actual sea surface height at any given time and place, and that which it would have if the ocean were at rest.
bottomT (degrees_C)	sea_water_potential_temperature_at_sea_floor is output directly from the native model; it is not interpolated. It means that bottom and the lowest wet point of the 3D temperature dataset could be different.
deptho_lev_interp	Deepest wet grid point of the Interpolated 3D fields.

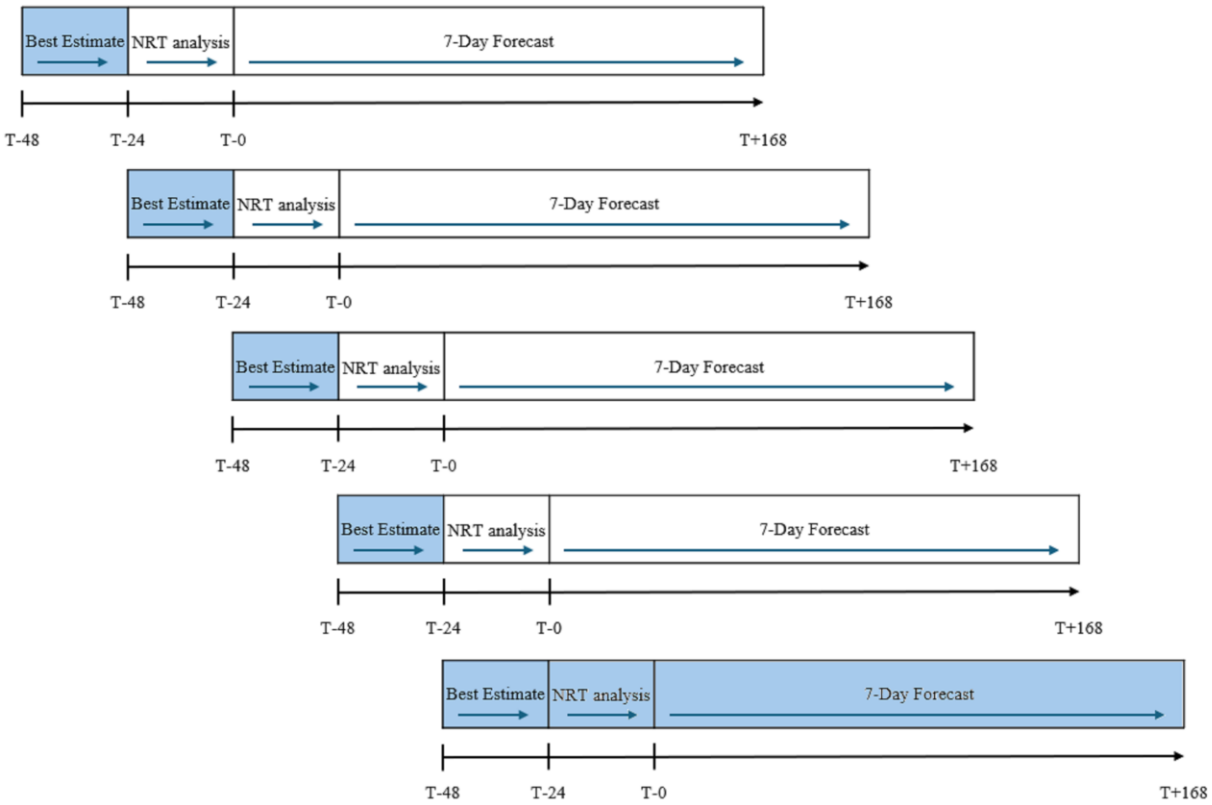
e) Processing information

Product disseminating schedule:

Type of product (forecast/ hindcast)	Forecast run (base time)	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
Forecast	00 UTC	0 to 168	until 12 UTC	Every day
NRT - Analysis	00 UTC	-24 to 0	until 12 UTC	Every day
Best estimate - Analysis	00 UTC	-48 to -24	until 12 UTC	Every day

f) Catalogue updating

The time coverage of the catalogue data spans from two years before present to present plus 7 days of forecast.



Each day the [T-48h, T-24h] and [T-24h, T+00h] analyses are provided along with 7 days forecast for this product (with the first day of the forecast being for the day of production).
The delivery usually happens between 6:40 AM and 10:00 AM, a couple of hours before the target delivery time. The previous day 'NRT analysis' and '7-day of forecast' are deleted at the beginning of this process. The catalogue data from 2-year before present to present minus 24 hours is taken from T-48h to T-24h of each run (Best Estimate – Analysis).

g) Time averaging

Dataset frequency	Time averaging	Forecast lead time
quarter hourly instantaneous values; time steps are in UTC	No temporal averaging or interpolation performed.	7-day (168-hr)
hourly instantaneous values; time steps are in UTC	No temporal averaging or interpolation performed.	
daily mean values	Values are calculated as means of 25-hour instantaneous values, starting at midnight (UTC) and finishing on the following midnight to remove both diurnal and tidal cycles.	

h) Product use and impact

Specialised marine physics analysis and forecast product for the Northwest Shelf region. Uses include marine navigation, emergency response (search and rescue/pollution), coastal flood mitigation, health and ecosystems monitoring, energy and net zero. Applications are developed for government policy, industrial decision making and academic research. The NWS products provide high resolution data and forecasts for ocean state in the tidally dominated waters of the Northwest Shelf and are designed to provide state of the art information specific to this region.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1) FILE SIZE

NORTHWESTSHELF_ANALYSIS_FORECAST_PHY_004_013		
DATASET NAME	**NAME OF FILE**	**DIMENSION [MB]**
cmems_mod_nws_phy-bottomt_anfc_1.5km-2D_P1D-m	metoffice_foam1_amm15_NWS_BED_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	0.9
cmems_mod_nws_phy-cur_anfc_1.5km-3D_P1D-m	metoffice_foam1_amm15_NWS_CUR_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	50
cmems_mod_nws_phy-mld_anfc_1.5km-2D_P1D-m	metoffice_foam1_amm15_NWS_MLD_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	0.8
cmems_mod_nws_phy-sal_anfc_1.5km-3D_P1D-m	metoffice_foam1_amm15_NWS_SAL_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	13
cmems_mod_nws_phy-tem_anfc_1.5km-3D_P1D-m	metoffice_foam1_amm15_NWS_TEM_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	16
cmems_mod_nws_phy-bottomt_anfc_1.5km-2D_PT1H-i	metoffice_foam1_amm15_NWS_BED_b{{yyyymmdd}}_hi{{yyyymmdd}}.nc	21
cmems_mod_nws_phy-cur_anfc_1.5km-3D_PT1H-i	metoffice_foam1_amm15_NWS_CUR_b{{yyyymmdd}}_hi{{yyyymmdd}}.nc	1331
cmems_mod_nws_phy-sal_anfc_1.5km-3D_PT1H-i	metoffice_foam1_amm15_NWS_SAL_b{{yyyymmdd}}_hi{{yyyymmdd}}.nc	305
cmems_mod_nws_phy-ssh_anfc_1.5km-	metoffice_foam1_amm15_NWS_SSH_b{{yyyymmdd}}_hi{{yyyymmdd}}.nc	13

2D_PT1H-i		
cmems_mod_nws_phy-tem_anfc_1.5km-3D_PT1H-i	metoffice_foam1_amm15_NWS_TEM_b{{yyyymmdd}}_hi{{yyyymmdd}}.nc	376
cmems_mod_nws_phy-cur_anfc_1.5km-2D_PT1H-i	metoffice_foam1_amm15_NWS_SSC_b{{yyyymmdd}}_hi{{yyyymmdd}}.nc	37
cmems_mod_nws_phy-mld_anfc_1.5km-2D_PT1H-i	metoffice_foam1_amm15_NWS_MLD_b{{yyyymmdd}}_hi{{yyyymmdd}}.nc	19
cmems_mod_nws_phy-sss_anfc_1.5km-2D_PT1H-i	metoffice_foam1_amm15_NWS_SSS_b{{yyyymmdd}}_hi{{yyyymmdd}}.nc	18
cmems_mod_nws_phy-sst_anfc_1.5km-2D_PT1H-i	metoffice_foam1_amm15_NWS_SST_b{{yyyymmdd}}_hi{{yyyymmdd}}.nc	20
cmems_mod_nws_phy-cur_anfc_1.5km-2D_PT15M-i	metoffice_foam1_amm15_NWS_SSC_b{{yyyymmdd}}_qh{{yyyymmdd}}.nc	147
cmems_mod_nws_phy-ssh_anfc_1.5km-2D_PT15M-i	metoffice_foam1_amm15_NWS_SSH_b{{yyyymmdd}}_qh{{yyyymmdd}}.nc	50
cmems_mod_nws_phy_anfc_1.5km_static	NWS-MFC_004_013_static.nc	161

4.2) STRUCTURE AND SEMANTIC OF NetCDF MAP FILES

An examples of the header of output NetCDF files is inserted in **annex**

5) REFERENCES

Quality Information Document
https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-013.pdf

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6) ANNEX

```
ncdump -h metoffice_foam1_amm15_NWS_TEM_b20250526_dm20250524.nc
netcdf metoffice_foam1_amm15_NWS_TEM_b20250526_dm20250524 {
dimensions:
        depth = 33 ;
        time = UNLIMITED ; // (1 currently)
        lat = 1240 ;
        lon = 958 ;
variables:
        double time(time) ;
                time:axis = "T" ;
                time:standard_name = "time" ;
                time:long_name = "Validity time" ;
                time:calendar = "Gregorian" ;
                time:units = "seconds since 1970-01-01T00:00:00Z" ;
        float lat(lat) ;
                lat:axis = "Y" ;
                lat:standard_name = "latitude" ;
                lat:units = "degrees_north" ;
                lat:nav_model = "grid_T" ;
                lat:cell_methods = "longitude: mean" ;
                lat:long_name = "latitude" ;
        float lon(lon) ;
                lon:axis = "X" ;
                lon:standard_name = "longitude" ;
                lon:units = "degrees_east" ;
                lon:nav_model = "grid_T" ;
                lon:cell_methods = "latitude: mean" ;
                lon:long_name = "longitude" ;
        short thetao(time, depth, lat, lon) ;
                thetao:_FillValue = -32768s ;
                thetao:missing_value = -32768s ;
                thetao:scale_factor = 0.001 ;
                thetao:add_offset = 25. ;
                thetao:standard_name = "sea_water_potential_temperature" ;
                thetao:long_name = "Sea Water Potential Temperature" ;
                thetao:units = "degrees_C" ;
                thetao:valid_min = -30000s ;
                thetao:valid_max = 30000s ;
        float depth(depth) ;
                depth:axis = "Z" ;
                depth:standard_name = "depth" ;
                depth:units = "m" ;
                depth:positive = "down" ;
                depth:long_name = "depth" ;

// global attributes:
                :title = "daily-mean potential temperature (3D)" ;
                :Conventions = "CF-1.7" ;
                :source = "PS-OS 47, AMM-FOAM 1.5 km (tidal) NEMO v3.6_WAVEWATCH-III
v4.18_NEMOVAR v6" ;
                :contact = "servicedesk.cmems@mercator-ocean.eu" ;
```

```
                :netcdf-version-id = "netCDF-4" ;
                :references = "http://marine.copernicus.eu/" ;
                :creation_date = "2025-05-26T06:27:51Z" ;
                :institution = "UK Met Office" ;
                :product = "NORTHWESTSHELF_ANALYSIS_FORECAST_PHY_004_013" ;
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :licence = "http://marine.copernicus.eu/services-portfolio/service-commitments-and-
licence/" ;
                :history = "" ;
                :forcing_data_source = "UKMO Unified Model (MetUM) Atmospheric model
Coupled-NWP (applied as direct fluxes); UKMO global ORCA12 ocean model; UKMO GLOWave;
CMEMS-BALTIC; climatological rivers" ;
}
```